PALIKA CHARITHA

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RESEARCH SUMMARY

Ph.D. researcher at IIT Madras working at the intersection of Reinforcement Learning (RL) and Computational Neuroscience. Work focuses on modeling brain-inspired decision-making and on designing closed-loop therapeutic control systems for neurological disorders using both deep RL and offline RL, with an emphasis on sequential decision-making under delayed rewards and uncertainty.

Expertise: Deep & Offline Reinforcement Learning, Sequential Decision-Making, Computational Neuroscience, Closed-Loop Neural Control, Multi-Agent RL, Generative Modeling (Flow Models, GANs)

EDUCATIONAL QUALIFICATIONS

Ph.D, Computational Neuroscience — IIT Madras *(Ongoing)* — CGPA: 9.48
M.Tech, Clinical Engineering — IIT Madras *(2020)* — CGPA: 9.82
B.Tech, Biotechnology — NIT Warangal *(2018)* — CGPA: 8.57

ONGOING RESEARCH

Adaptive Deep Brain Stimulation (DBS) using Proximal Policy Optimization (PPO) based Reinforcement Learning.

- Formulated DBS control as a sequential decision-making problem over neural state trajectories to suppress the diseased activity in Parkinson's disease.
- Designed a closed-loop PPO-based controller to optimize stimulation parameters such as pulse timing and pulse amplitude.

Offline Reinforcement Learning for L-DOPA Medication Optimization for Parkinson's using PPMI Dataset.

- Formulated medication dosing as an MDP using clinical patient data obtained from the PPMI dataset
- Developed an offline Q-learning framework for safe policy learning under clinical and safety constraints.

Deep Reinforcement Learning for Diet Optimization and Glucose Regulation

- Designed a sequential policy learning framework (LSTM + PPO) for personalized glucose regulation.
- Addressed delayed rewards and temporal dependencies using eligibility traces for improved credit assignment.

Cooperative Multi-Agent Modeling of Gut–Brain Energy Regulation using MAPPO

- Modeled gut–brain energy regulation as a cooperative multi-agent RL problem in a custom Gym environment.
- Learned coordinated decentralized policies using MAPPO for maintaining energy homeostasis under stochastic dynamics.

PUBLICATIONS AND RESEARCH OUTPUT:

Manuscripts (Under Review)

Charitha Palika, Aniket Khan, V. Srinivasa Chakravarthy; **NutriRL: A Benchmark for Nutritional Regulation under Delayed State Transitions. (Under review at Reinforcement Learning Conference (RLC) 2026)**

- Introduced a novel RL benchmark with multi-channel delayed state dynamics, moving beyond conventional reward-delay formulations to capture realistic physiological feedback
- Systematically analyzed algorithm sensitivity to delay structure, showing that delayed state evolution, not just reward sparsity, drives credit assignment difficulty

Charitha Palika, Nurani Rajagopal Rohan, Aniket Malviya, V. Srinivasa Chakravarthy; **Harnessing Deep Brain Stimulation (DBS) to Improve Exploration and Optimize Decision-Making through STN–GPe Desynchronization. (Under review at PLOS Computational Biology)**

- Designed an RL-based computational framework for decision-making in Parkinson's disease, incorporating a meta-learning rule linking dopaminergic activity to exploration dynamics.
- Demonstrated that adaptive DBS improves decision-making performance specifically on non-stationary bandits, and proposed a novel stimulation protocol outperforming standard DBS.

Conference Socials & Posters

Saketh Ganti, Charitha Palika, V. Srinivasa Chakravarthy; **NeuroAI: From Neurons to Transfomers - NeurIPS 2025: Socials.** link

- Co-organized the *NeuroAI: From Neurons to Transformers* social at NeurIPS 2025, an interdisciplinary forum fostering dialogue between neuroscience and AI on brain-inspired models, learning systems, and education

Charitha Palika, Aniket Khan, V. Srinivasa Chakravarthy; **A Reinforcement Learning Model of Dopamine-Mediated Appetite Regulation in Health, Parkinson's Disease, and Dopaminergic Treatment, International Conference on Alzheimer's and Parkinson's Diseases - AD/PD 2026** link

- Designed a custom OpenAI Gym environment for modeling appetite regulation and trained a Monte Carlo Actor Critic agent to learn normative and diseased behavioral policies.

Charitha Palika, Sandeep Satyanandan Nair, V. Srinivasa Chakravarthy; **Modeling the basal ganglia for cognitive performance in healthy control, parkinson's disease condition, and DBS-treated patients, International Conference on Alzheimer's and Parkinson's Diseases - AD/PD 2024** link

- Designed an RL-based computational model of the basal ganglia to study cognitive performance in healthy, Parkinson's, and DBS-treated conditions.

Charitha Palika, Jayanth Sharma, Nurani Rajagopal Rohan, Sandeep Satyanandan Nair, V. Srinivasa Chakravarthy; **Modeling the exploratory dynamics of the Indirect Pathway in the Basal Ganglia using a network of Chaotic Attractors, Society for Neuroscience 2024 (SFN24)** link

- Developed an RL model of the basal ganglia indirect pathway with exploratory dynamics driven by chaotic attractors; presented at SfN 2024.

PROJECT DETAILS

Selective Focus: Multi-Agent Reinforcement Learning with Dynamic Attention in Partially Observable Continuous Action Spaces. (DA7400 course project) link

- Designed MARL framework using Graph Attention Networks under partial observability.
- Improved coordination performance in continuous control environments (Predator-Prey, Battlefield) using **graph attention networks**, outperforming the baseline MADDPG method.

Attention-Enhanced Real NVP for High-Dimensional Density Estimation : (EE6180 - Advanced AI: Course Project) link

- Extended flow-based models with **attention-based coupling (CBAM)** for image generation.
- Improved density estimation and generative performance.

GRU Networks for Robotic Control: Stability and Fixed Point Analysis in a 2-Link Arm (Neuromatch NeuroAI project) link

- Trained recurrent models for 2-link arm control (MuJoCo).
- Performed fixed-point analysis for interpretability and stability.

Cross-Domain Image Translation: Implementing RGB to Thermal Mapping of traffic signal images. (EE5179 course project) link

- Implemented RGB-to-thermal image translation using Pix2Pix GAN, U-Net++, and MANet
- Compared model performance to evaluate the accuracy of thermal image generation from RGB inputs

COURSEWORK

Deep learning for imaging (EE5179)	Reinforcement learning (CS6700)
Recent advancements in Reinforcement learning (DA7400)	Advanced Topics in Artificial Intelligence (EE6180)
Modern Computer Vision (EE5178)	Computational neuroscience (BT6270)

SKILLS

Python, PyTorch, OpenCV, Scikit-learn, Gymnasium, MuJoCo, TorchRL, PettingZoo

ACHIEVEMENTS

- **Institute Merit Prize (M.Tech) for the best academic record** in the Biotechnology Department, IIT Madras
- **American Express Award, Dr. S.S. Srikanta Prize, P K Narayana Iyer Prize for academic excellence** in IIT Madras (M.Tech).
- **Buti Foundation Gold Medal Award (IIT Madras)** for best academic record (women) amongst Dual Degree / M.Tech / M.Sc / BS / MS Program for the session 2018-20.

COMPETITIVE EXAMINATIONS

- GATE Biotechnology (2018): **AIR 15**
- Joint CSIR-UGC JRF & Eligibility for Lectureship (NET) Lifesciences (2018): **AIR 113**